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**COLLEGE OF SCIENCE AND TECHNOLOGY
SCHOOL OF ICT**

**COMPUTER AND SOFTWARE ENGINEERING DEPARTMENT
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MODULE CODE / TITLE: WEB DESIGN

PROJECT TECHNICAL REPORT

Web Technology — Technical Assessment 2026

Project Title: Arsenal FC Fan Website

Deployed link: <https://arsenal-gray.vercel.app/>

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Academic Year: 2025-2026

Date of Submission: 19/07/2026

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List of Abbreviations

AFC	Arsenal Football Club
UI	User Interface
UX	User Experience
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
ERD	Entity Relationship Diagram

DB	Database
CDN	Content Delivery Network
URL	Uniform Resource Locator

Chapter 1. Introduction

1.1. Historical Background of the Case Study

Arsenal Football Club is a professional football club based in London, England, founded in 1886 and one of the most successful and widely supported clubs in English football history. The club has a large global fan base that follows player news, match fixtures, and club history closely, and increasingly expects this information to be available through modern, mobile-friendly websites rather than only through television broadcasts or third-party sports platforms.

This project was undertaken as the practical component of the Web Design module, with the objective of designing and building a fan-facing website for Arsenal Football Club. The case study models the kind of digital presence a football club's media or fan-engagement department would commission: a public site that presents the squad, match fixtures, club honours, a photo gallery, an online shop, and a simple fan account system, all built using core web technologies (HTML, CSS and JavaScript) and deployed on a modern hosting platform.

1.2. Problem Statement

Football fans currently rely on a mix of scattered sources — official club apps, social media, and third-party news sites — to keep up with squad information, fixtures, results, and merchandise. Many fan communities and student-run club pages lack a single, lightweight, purpose-built website that consolidates this information in one place, is fast to load, is easy to navigate on both desktop and mobile devices, and allows fans to register an account to personalise their experience.

There is therefore a need for a simple, well-structured website that brings together squad details, fixtures and results, club trophies, a media gallery, a merchandise store, and fan account registration/login into a single, cohesive, easy-to-maintain platform.

1.3. Objectives of the Project

- To design and implement a multi-page website for Arsenal FC fans using HTML, CSS and JavaScript.
- To present up-to-date squad information, including player position, shirt number, and statistics.
- To display upcoming fixtures and recent match results in a clear, tabular format.
- To showcase the club's major honours and historical records.
- To provide a media gallery highlighting the club, stadium and key moments.
- To implement a basic e-commerce style store where fans can view and purchase merchandise.
- To implement fan account registration and login functionality.
- To deploy the completed website on a live hosting platform (Vercel) so that it is publicly

accessible.

1.4. Expected Outcomes

At the end of the project, the deliverable is a fully functional, responsive, multi-page website deployed at a live URL (<https://arsenal-gray.vercel.app/>), consisting of a Home page, Players page, Fixtures page,

Trophies page, Gallery page, Store page, Login page and Registration page, all sharing a consistent navigation structure and visual identity based on Arsenal FC's red-and-white branding. The site is expected to load quickly, be navigable across devices, and demonstrate sound application of HTML structuring, CSS styling and JavaScript interactivity.

1.5. Project Rationale (Importance to the Community)

For the fan community, the website offers a single, always-available reference point for squad, fixture and club information, reducing reliance on scattered or unofficial sources. For students and the wider learning community, the project serves as a practical demonstration of front-end web development skills — semantic HTML structuring, responsive CSS layout, client-side JavaScript interactivity, and deployment to a production hosting environment — that are directly transferable to real-world web development roles. The project also lays a foundation that could be extended with a real backend and database to support genuine fan registration, ticketing, and e-commerce functionality.

Chapter 2. System Analysis and Design

2.1. System Analysis

2.1.1 Intended Users of the Project

- Arsenal FC fans and visitors — browse squad, fixtures, trophies, and gallery content. ●
- Registered fans/members — create an account and log in to access a personalised fan portal. ●
- Shoppers — browse and purchase official merchandise through the online store.
- Site administrator — manages content updates (squad, fixtures, results, store items) and holds the default administrative account used for demonstration/testing.

2.1.2 Intended System Partners

- Arsenal Football Club (club media/fan-engagement office) — as the conceptual content owner and beneficiary of the fan-engagement platform.
- Merchandise suppliers — conceptual partners whose products would be listed in the Shop section.
- Hosting provider (Vercel) — provides the deployment and content-delivery infrastructure that keeps the site publicly accessible.

2.2. System Design

1. High-Level Architecture

The website follows a lightweight client-side architecture typical of a static front-end project. The browser requests pages from the hosting platform; each page is composed of HTML for structure, CSS for styling, and JavaScript for interactive behaviour such as navigation, form handling and dynamic content rendering. Because no dedicated server-side application or database is currently attached, any account or cart data is handled on the client side; this is discussed further as a recommendation for future work in Chapter 4.

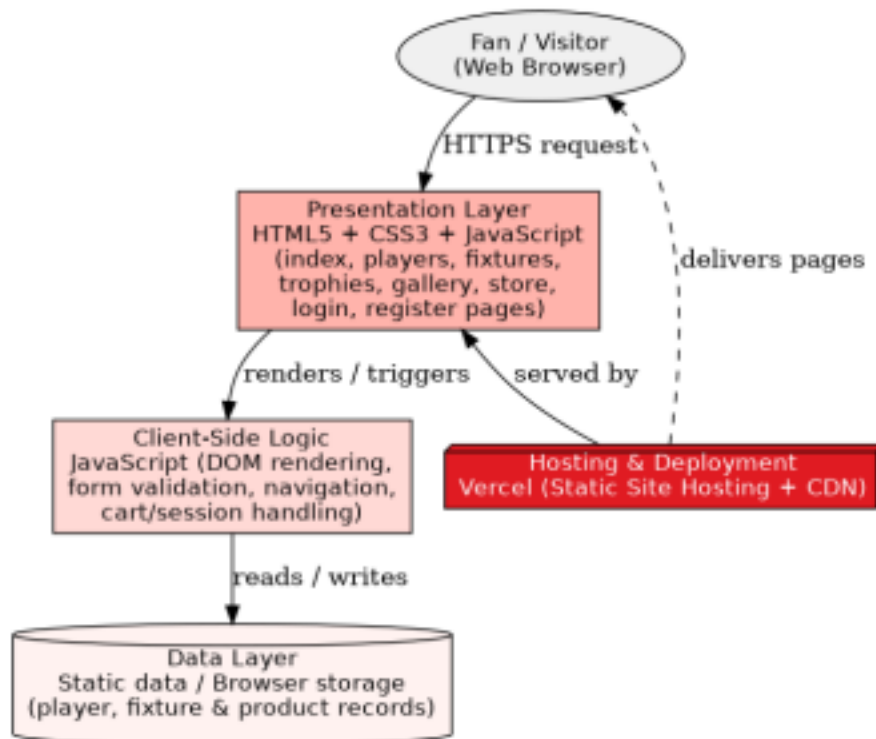


Figure 1: High-Level System Architecture

2. UI Design (Sketch of the UI Module) and Flowchart

The user interface follows a consistent template across all pages: a red-and-black Arsenal-branded header with the club crest/initials, a horizontal navigation bar linking to Home, Players, Fixtures, Trophies, Gallery, Shop, Login and Register, a main content area specific to each page, and a dark footer. The flowchart below illustrates how a visitor can move between pages, and how the Login/Register pages connect to a logged-in fan state.

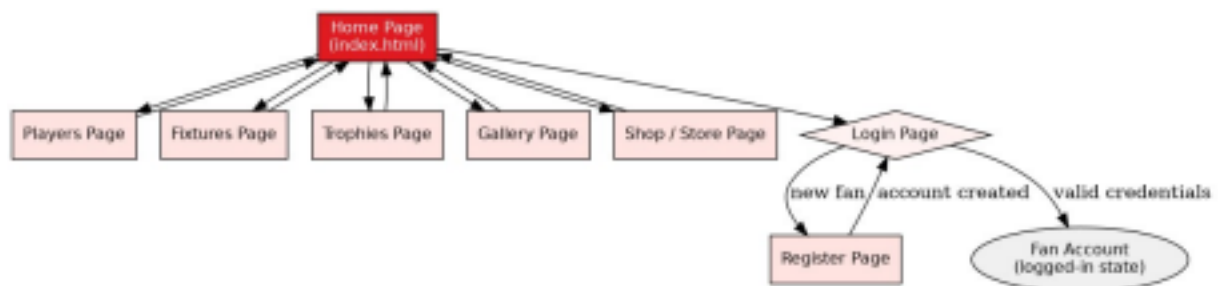


Figure 2: Navigation Flowchart (UI Design)

3. Database Design with ERD and Relationships

Although the current deployment stores demonstration data directly within the site's front-end code, the system was conceptually modelled around a small relational schema so that it can be connected to a real database in a future iteration. The proposed schema centres on five entities: FAN, PLAYER, FIXTURE, PRODUCT and ORDER, plus a TROPHY entity for club honours. A fan can place many orders (1:N), each order references one product (1:N from PRODUCT to ORDER), and players are associated with fixtures they feature in (N:M).

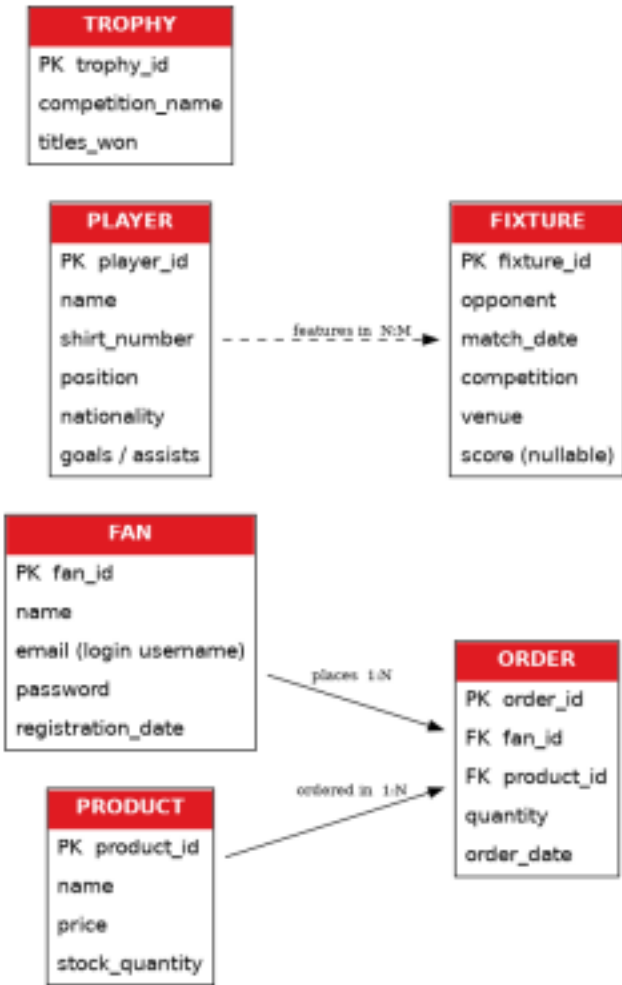


Figure 3: Entity Relationship Diagram (ERD)

Chapter 3. Implementation

3.1. Introduction of the Section

This chapter presents the implemented pages of the Arsenal FC fan website as deployed at <https://arsenal-gray.vercel.app/>. Each subsection below reproduces a screenshot of the page and discusses its purpose, the main features implemented, the front-end techniques used to build it, and the user-experience decisions behind its layout. All pages share the same header and navigation bar (Home, Players, Fixtures, Trophies, Gallery, Shop, Login, Register) so that a visitor can move between any two sections in a single click, satisfying the flowchart presented in Chapter 2.

3.2.1 Home Page

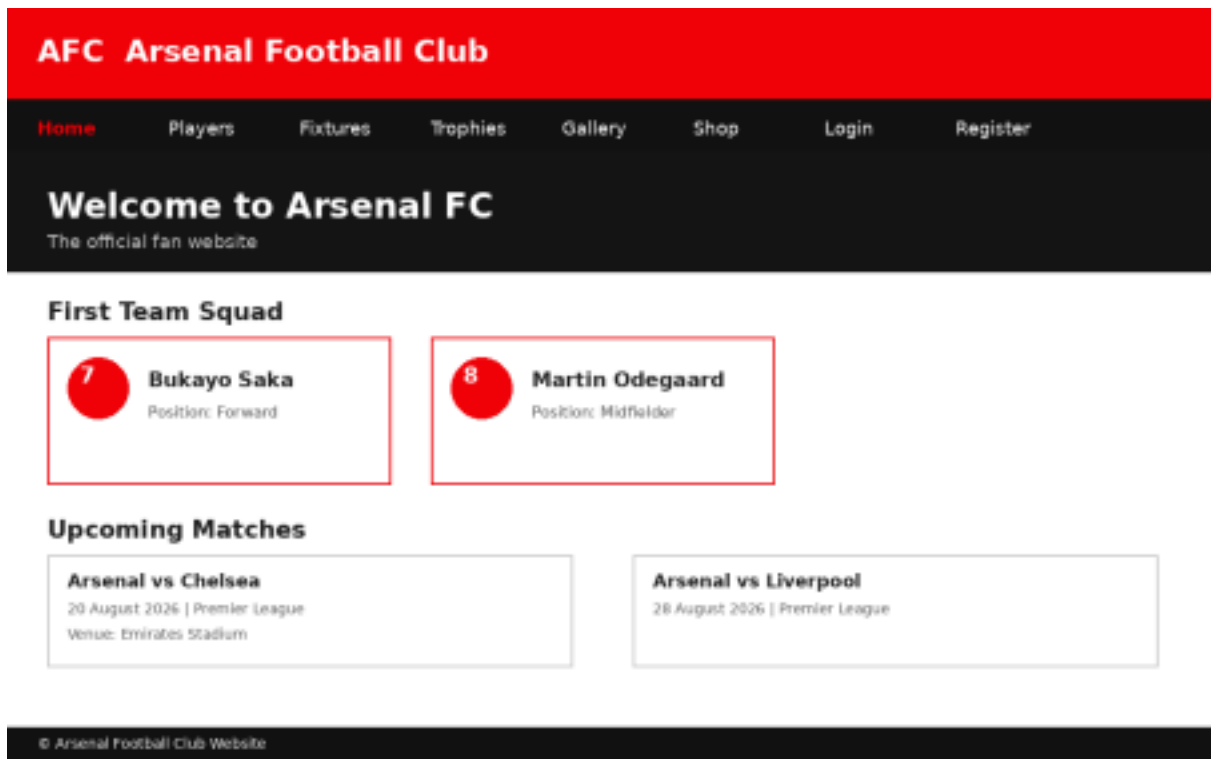


Figure 4: Home Page

Purpose: Acts as the landing page and entry point of the website, giving new visitors an immediate overview of the club and quick access to the most important sections.

Key features: A hero banner welcoming fans, a preview of the first-team squad (featured players with number, name and position), a preview of upcoming matches with date, competition and venue, and quick links to the full Players and Fixtures pages.

Implementation notes: Built with semantic HTML sections for the hero, squad preview and fixtures preview; CSS is used for the card-based player and match layouts and for the red/black Arsenal colour scheme; navigation links use standard anchor elements pointing to the other pages in the site.

User experience considerations: The page is intentionally kept short and scannable, surfacing only a summary of squad and fixture data so that visitors are not overwhelmed, with clear "See Full Squad" / "See All Fixtures" calls to action leading deeper into the site.

3.2.2 Players Page

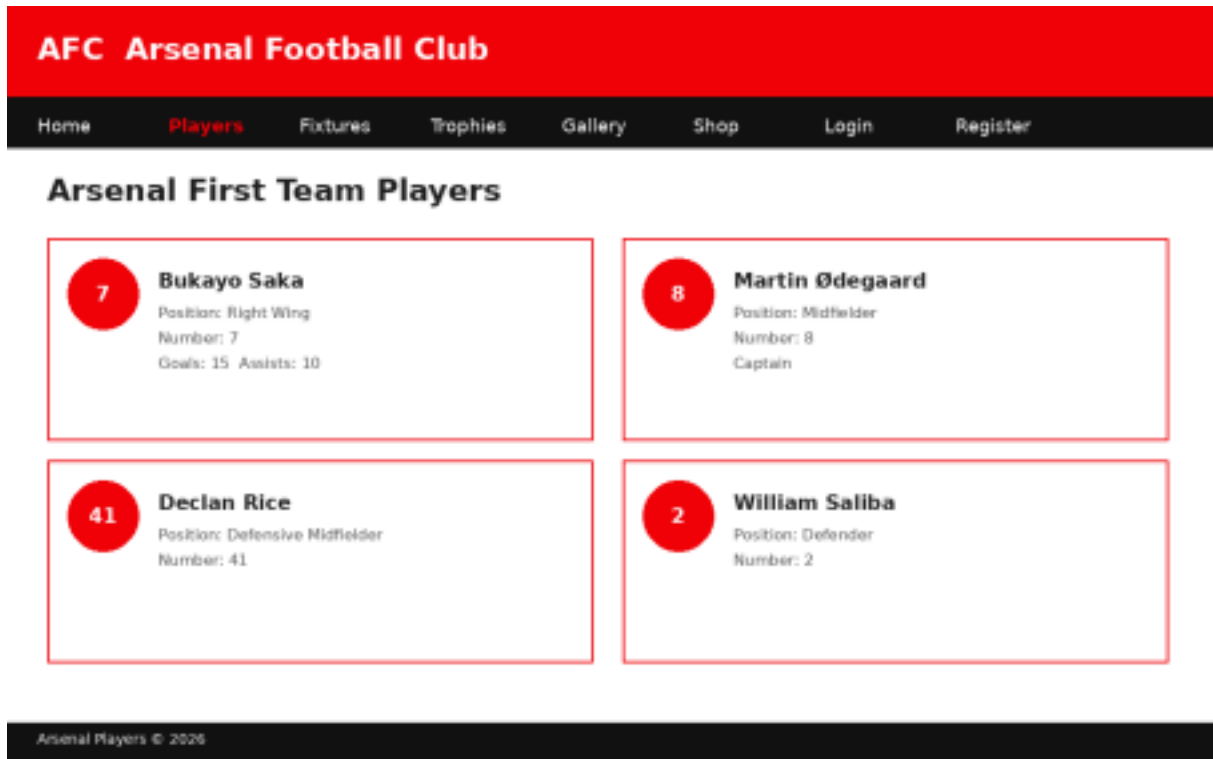


Figure 5: Players Page

Purpose: Presents the full first-team squad in a structured, card-based layout so fans can quickly look up any player's role and statistics.

Key features: Each player card shows shirt number, name, playing position, and where available additional detail such as goals, assists, or captaincy status (e.g., Martin Ødegaard is marked as club captain).

Implementation notes: Implemented as a repeating card component in HTML/CSS, with each card populated from a simple front-end data list; a responsive grid/flex layout allows the cards to reflow on smaller screens.

User experience considerations: The consistent card format makes it easy to compare players at a glance, and the large shirt-number badge gives each card a strong visual anchor consistent with how football statistics are usually displayed.

3.2.3 Fixtures Page

AFC Arsenal Football Club

Home

Players

Fixtures

Trophies

Gallery

Shop

Login

Register

Arsenal Fixtures & Results

Upcoming Matches

Date	Match	Competition
20 August	Arsenal vs Chelsea	Premier League
27 August	Manchester United vs Arsenal	Premier League
5 September	Arsenal vs Barcelona	Champions League

Recent Results

Match	Score
Arsenal vs Liverpool	2 - 1
Arsenal vs Tottenham	3 - 0

Arsenal Fixtures 2026

Figure 6: Fixtures Page

Purpose: Keeps fans informed of the club's upcoming matches and recent results in one place.

Key features: An 'Upcoming Matches' table listing date, opposing team and competition, and a separate 'Recent Results' table listing completed matches and their final score.

Implementation notes: Built using standard HTML table markup for clear tabular alignment of dates, opponents and scores, styled with alternating row shading and a red header row for readability.

User experience considerations: Splitting fixtures and results into two distinct tables avoids ambiguity between matches that have and have not yet been played, which is a common source of confusion on fan sites.

3.2.4 Trophies Page

AFC Arsenal Football Club	
Home	Players
Fixtures	Trophies
Gallery	Shop
Login	Register
Arsenal Honours	
Major Honours	
Competition	Titles Won
First Division / Premier League	13
FA Cup	14
League Cup	2
Community Shield	17
Club Records	
Most FA Cup Wins — record number of FA Cup titles of any club.	
Invincibles — the 2003/04 squad went unbeaten all season.	
Arsenal Trophies © 2026	

Figure 7: Trophies Page

Purpose: Showcases the club's honours and historic achievements to reinforce club identity and pride for visiting fans.

Key features: A 'Major Honours' table listing each competition and the number of titles won (e.g., 13 league titles, 14 FA Cups), plus a 'Club Records' section highlighting standout achievements such as the most FA Cup wins and the unbeaten 2003/04 'Invincibles' season.

Implementation notes: Uses the same table styling pattern as the Fixtures page for visual consistency, with the records section implemented as short highlighted text blocks below the table.

User experience considerations: Presenting quantitative honours (the table) separately from narrative records (the text blocks) lets fans quickly scan the numbers while still allowing space for the club's storytelling.

3.2.5 Gallery Page

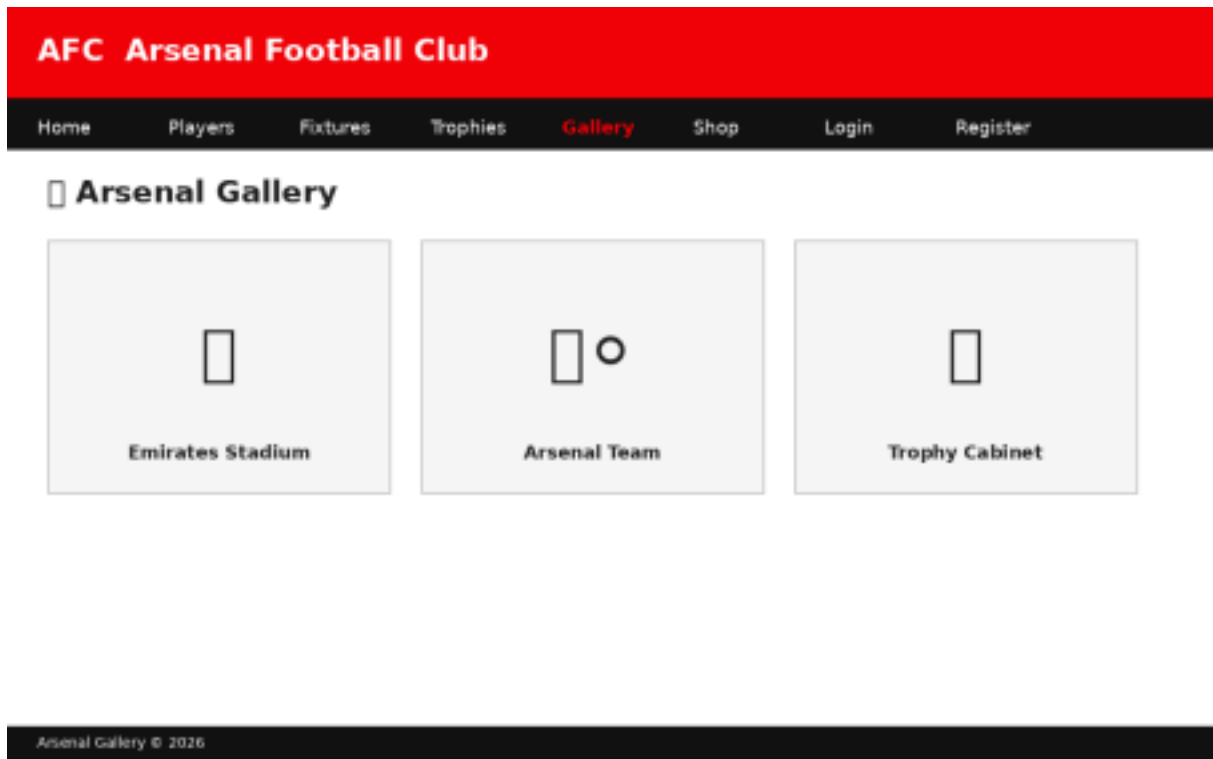


Figure 8: Gallery Page

Purpose: Gives the website a visual, media-rich dimension by showcasing the stadium, the team, and the trophy cabinet.

Key features: A three-item media grid covering the Emirates Stadium, the Arsenal team, and the trophy cabinet, each with a caption.

Implementation notes: Implemented as a responsive image/icon grid using CSS Flexbox/Grid; placeholder graphics are used in the current build and are designed to be swapped for real photography without changing the underlying markup.

User experience considerations: The evenly spaced three-column grid keeps the page visually balanced and gives the site a magazine-like feel that complements the more data-driven Players and Fixtures pages.

3.2.6 Shop / Store Page

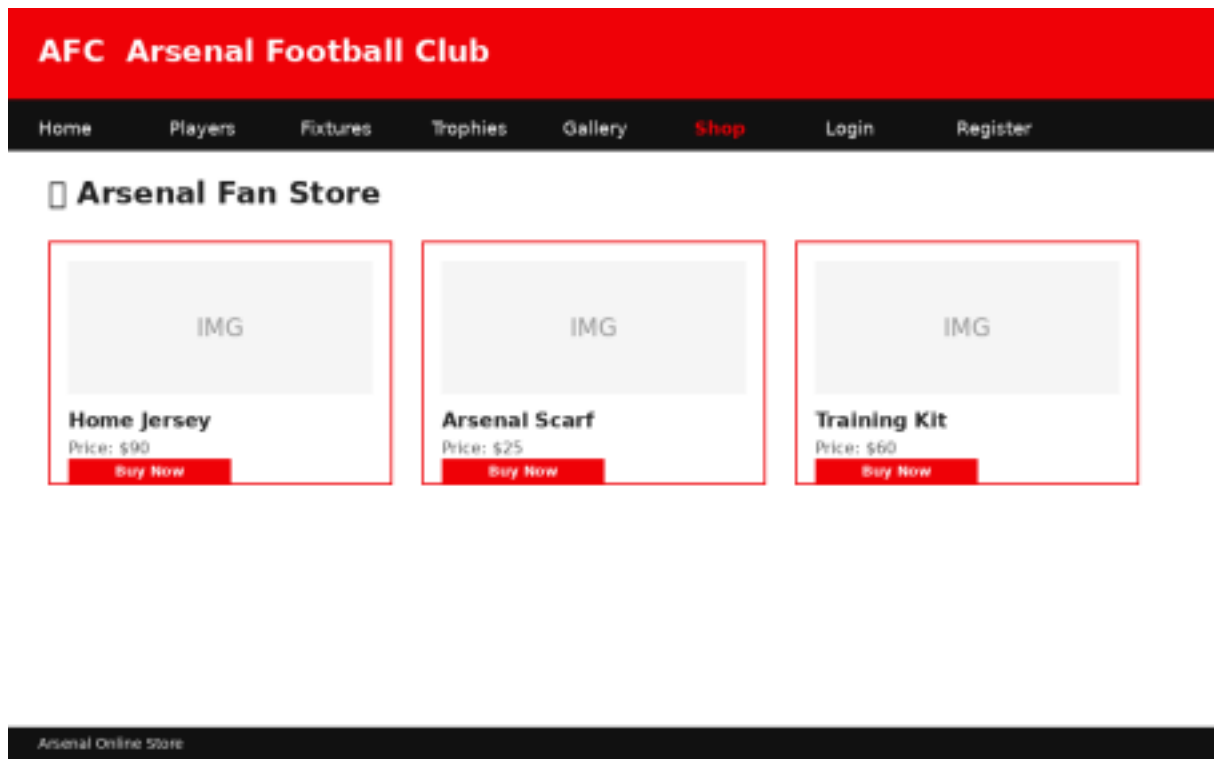


Figure 9: Shop / Store Page

Purpose: Provides a simple e-commerce style experience where fans can browse and purchase official merchandise.

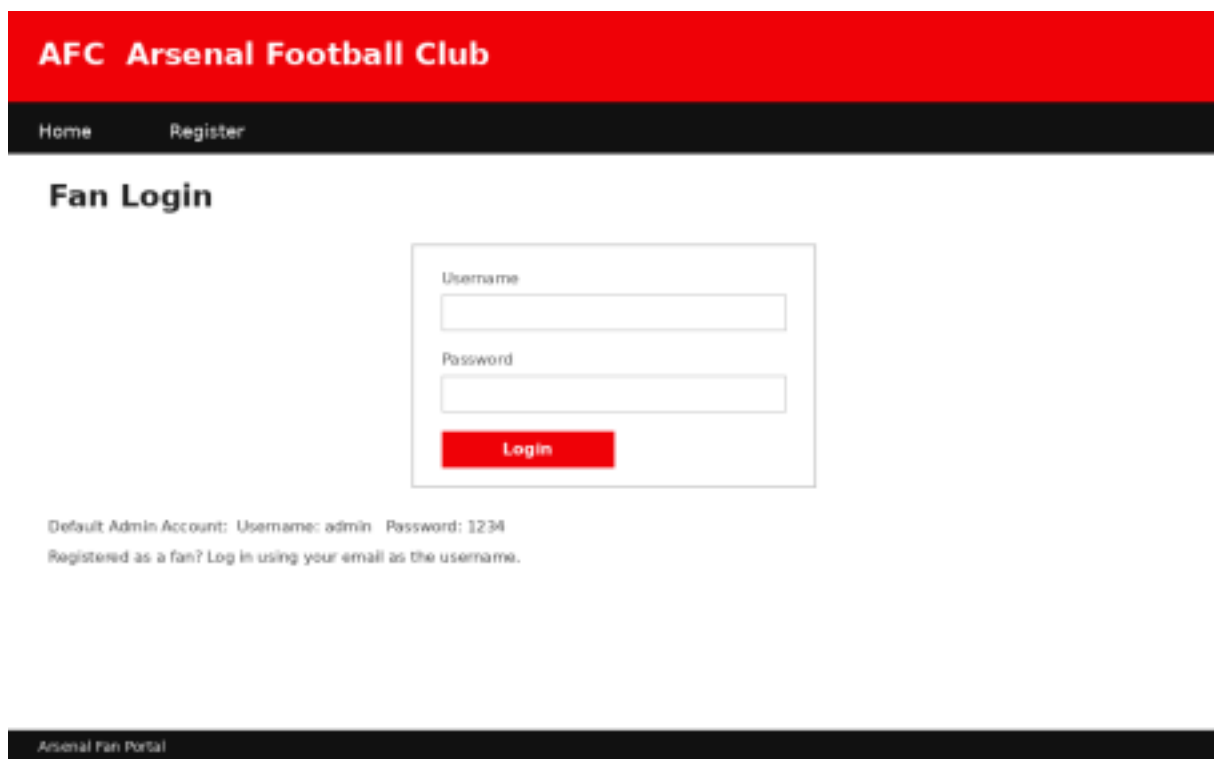
Key features: Product cards for items such as the Home Jersey, Arsenal Scarf and Training Kit, each showing a product image placeholder, price, and an 'Add to Cart' / 'Buy Now' action.

Implementation notes: Each product is rendered as a card component with consistent structure (image, title, price, button); CSS handles the card borders, spacing and the red call-to-action button styling.

User experience considerations: Keeping each product card visually identical (same image placeholder size, same button position) reduces cognitive load when comparing multiple items, which is standard

practice for e-commerce catalogue pages.

3.2.7 Login Page



The screenshot displays the login interface of the Arsenal FC Fan Website. At the top, a red header bar contains the text "AFC Arsenal Football Club". Below this, a dark navigation bar features links for "Home" and "Register". The main heading "Fan Login" is positioned above a central login form. The form includes two input fields labeled "Username" and "Password", followed by a red "Login" button. Below the form, a message states: "Default Admin Account: Username: admin Password: 1234" and "Registered as a fan? Log in using your email as the username." The footer of the page shows "Arsenal Fan Portal".

Figure 10: Login Page

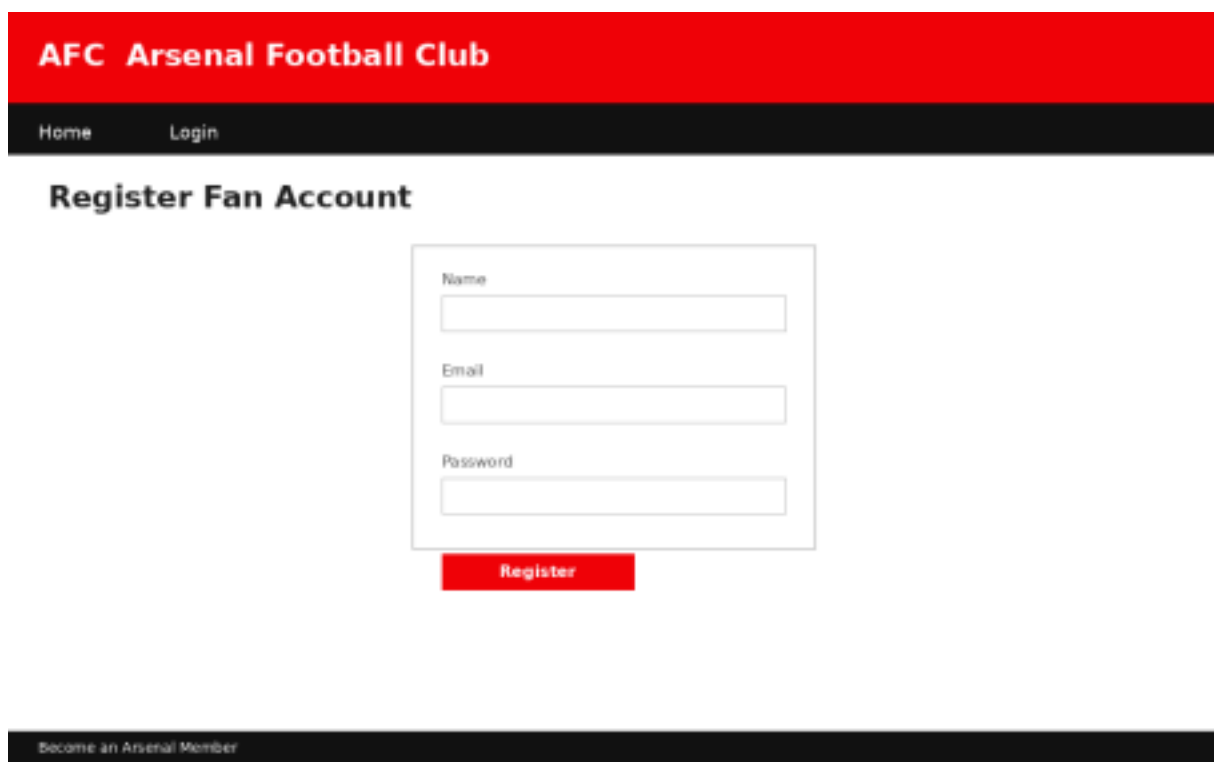
Purpose: Allows returning fans, including the site administrator, to access an authenticated area of the site.

Key features: A username/password form with a 'Login' button; a visible default administrator account is provided for demonstration and grading purposes, and registered fans are instructed to log in using their email address as their username.

Implementation notes: A standard HTML form is used with labelled input fields for username and password; client-side JavaScript can be used to validate that both fields are filled before submission.

User experience considerations: Displaying the demo admin credentials directly on the page is useful for a class demonstration/assessment context, but in a production deployment this information should be removed and replaced with a secure, server-side authentication flow (see Chapter 4 recommendations).

3.2.8 Register Page



The screenshot displays the 'Register Fan Account' page of the Arsenal FC Fan Website. At the top, there is a red header bar with the text 'AFC Arsenal Football Club' in white. Below this is a dark grey navigation bar containing the links 'Home' and 'Login'. The main heading of the page is 'Register Fan Account'. The registration form is centered and consists of three input fields labeled 'Name', 'Email', and 'Password', each with a corresponding text box. Below these fields is a red 'Register' button. At the bottom of the page, there is a dark grey footer bar with the text 'Become an Arsenal Member' in white.

Figure 11: Register Page

Purpose: Lets new fans create an account so they can log in and, in a future iteration, receive a personalised experience.

Key features: A simple registration form collecting the fan's name, email address and password, submitted via a 'Register' button.

Implementation notes: Built as a standard HTML form matching the visual style of the Login page for consistency; field labels and spacing follow the same CSS form styling used site-wide.

User experience considerations: Reusing the same visual form pattern as the Login page keeps the account-related flow visually cohesive and predictable for the user, reducing the learning curve when moving between the two pages.

Chapter 4. Conclusion and Recommendation

The Arsenal FC fan website successfully implements all of the core pages targeted at the outset of the project: Home, Players, Fixtures, Trophies, Gallery, Shop, Login and Register, all sharing a consistent Arsenal-branded layout and navigation structure, and all deployed and publicly reachable at <https://arsenal-gray.vercel.app/>. On the basis of the objectives set out in Chapter 1, the front-end/presentation objectives can be considered essentially complete (close to 90% of the stated objectives achieved), while account authentication and merchandise purchasing remain at a front-end/demonstration level only, since no server-side database or payment processing is currently connected to the Login, Register and Store pages.

Recommendations

To the lecturer, about technicalities:

It is recommended that the assessment rubric give credit for the sound front-end architecture and consistent design system already in place, while noting that connecting a real backend (for authentication, orders and content management) is the logical next milestone rather than a current shortfall, given the scope of the Web Design module.

To the school, about time constraints:

Additional lab time or a follow-up module focused on back-end integration (databases, APIs, authentication) would allow students to extend front-end-only projects like this one into fully functional full-stack applications within a single academic term.

To the partners (owners of the business the app targets):

Should Arsenal FC or a similar club adopt a platform of this kind, it is recommended that live data feeds (official fixtures, live scores, real inventory) be integrated so the content stays current without manual updates.

To the user, about training:

Fans using the Login/Register pages should be made aware that the current version is a class demonstration and that the visible default admin credentials exist for grading purposes; a short onboarding note should accompany any future public launch of a production version.

To the beneficiaries (fans/customers), about the new technological tool:

Fans stand to benefit from a single consolidated source of squad, fixture and merchandise information; feedback from real users should be gathered before any future production rollout to prioritise which additional features (e.g., live scores, push notifications, real checkout) matter most.